Panel Data Analysis

F. Candau (UPPA-TREE)

M1 ECO

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What is panel data?

- **Cross-section:** many units i, each observed **once**.
- **Time series: one** unit, many periods t (e.g. French GDP over time).
- **Panel (longitudinal):** the **same** units i followed over periods t — at least **two** dimensions.

Higher dimensions are common:

- exports between i and j, at time t, for product k;
- wages of workers i in sectors k, cities c, over years t.

The first problem: what the error hides

We want the causal effect β of x_{it} on y_{it}. Run a naive regression and everything we cannot measure lands in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} = \underbrace{f_i}_{\text{individual}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

- f_i — **unobserved individual** characteristics: ability, institutions, geography… fixed over time, differing across units.
- f_t — **time** effect: shocks hitting **all** units in period t (crises, world prices, technology).
- If f_i or f_t correlates with x_{it}, OLS is **biased** (omitted-variable bias).

The promise of panel data

Observing the *same* units over *several* periods lets us **sweep f_i and f_t out of the error** — identifying β even when they correlate with x.

The second problem: errors that move together

Problem 1 in one line: $\hat{\beta} = (X'X)^{-1}X'y$ is biased when $\mathbb{E}[\varepsilon \mid X] \neq 0$, and ε contains $f_i + f_t$. Sections 1–3 remove them by **transforming the data** before running OLS.

A second problem sits in the same error. Take one unit in two periods s and t:

$$\varepsilon_{is} = f_i + f_s + u_{is}, \qquad \varepsilon_{it} = f_i + f_t + u_{it}$$
$$\mathrm{Cov}(\varepsilon_{is}, \varepsilon_{it}) = \mathrm{Var}(f_i) + \mathrm{Cov}(u_{is}, u_{it}) \neq 0.$$

- The errors of the **same unit are correlated** across its periods. This does not bias $\hat{\beta}$; it changes its **variance**,

$$\mathrm{Var}(\hat{\beta}) = (X'X)^{-1}\, X'\Omega X\, (X'X)^{-1}, \qquad \Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X].$$

- Default software assumes $\Omega = \sigma^2 I$, independent errors. Here that is false, and the printed standard errors are **too small**.
- The section *Clustered standard errors* gives the correct Ω.

Balanced vs. unbalanced panels

- **Balanced:** every unit i is observed in **every** period t — exactly $N \times T$ rows (a full rectangle).
- **Unbalanced:** some unit–period cells are **missing** (units enter/exit, attrition) — fewer than $N \times T$ rows.

Why it matters

A couple of results below — notably that the **between** estimator removes the time effect — hold cleanly **only for balanced panels**. Real data are often unbalanced; we flag where it bites.

Roadmap

1. Panel data: dimensions and the model.
2. Estimators that **ignore** heterogeneity: pooled OLS and between.
3. The **within (fixed-effects)** estimator.
4. **Two-way fixed effects**.
5. Standard errors: what Ω is (clustering).
6. Application: temperature and growth (Dell et al. 2012).

We deliberately skip random effects and the Hausman test. Random effects requires $\mathrm{Cov}(x_{it}, f_i) = 0$ — the very assumption panel data let us **avoid** — so it is rarely defensible in modern applied work.

Section 1

Naive estimators

Pooled OLS

Treat the panel as one big cross-section, leaving **both** unobserved effects in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} = f_i + f_t + u_{it}.$$

- Removes **neither** f_i nor f_t from the error.
- **Consistent only if** $\mathrm{Cov}(x_{it}, f_i) = 0$ **and** $\mathrm{Cov}(x_{it}, f_t) = 0$ — usually too strong.
- ε_{it} shares f_i across all periods, so a unit's errors are **correlated with each other**: the second problem, treated in the section *Clustered standard errors*.

The bias, exactly

With a single regressor and exogenous idiosyncratic error ($\mathrm{Cov}(x_{it}, u_{it}) = 0$), $\operatorname{plim} \hat{\beta}_{\text{pooled}} = \beta + \dfrac{\mathrm{Cov}(x_{it},\, f_i + f_t)}{\mathrm{Var}(x_{it})}$ — nonzero the moment x moves with *either* unobserved effect. The transformations below kill that covariance.

The between estimator

Run OLS of $\bar{y}_i$ on $\bar{x}_i$ — a *between* regression on the **unit means** (N data points, one per unit) with error $\bar{\varepsilon}_i$:

$$\bar{y}_i = \alpha + \beta\,\bar{x}_i + \bar{\varepsilon}_i, \qquad \hat{\beta}_B = \frac{\sum_i (\bar{x}_i - \bar{x})(\bar{y}_i - \bar{y})}{\sum_i (\bar{x}_i - \bar{x})^2}.$$

What sits in that error? Average the true model over t (f_i is constant, so it stays). In a **balanced panel** the time effects average to the same grand mean for every unit:

$$\bar{\varepsilon}_i = f_i + \bar{f} + \bar{u}_i, \qquad \bar{f} = \tfrac{1}{T}\sum_t f_t.$$

- That $\bar{f}$ is the **same for all units** $\to$ a constant the intercept absorbs: **between sweeps out** f_t (its quiet advantage).
- f_i is **still in the error** $\to$ **biased if** $\mathrm{Cov}(\bar{x}_i, f_i) \neq 0$.
- Uses **only between** variation; a benchmark, rarely final. *(Unbalanced panel: the average differs across units and need not wash out.)*

What does each estimator remove from the error?

The error hides two unobserved effects, f_i and f_t. Each estimator is OLS on transformed data, and the transformation decides **which effect leaves the error**:

	data fed to OLS	removes f_i?	removes f_t?	left biased by
Pooled	raw rows x_{it}	no	no	f_i and f_t
Between	unit means $\bar{x}_i$	no	yes	f_i
Within	$x_{it} - \bar{x}_i$	yes	no	f_t
TWFE	$x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$	yes	yes	—

- **Between** averages over time $\rightarrow$ kills f_t, keeps f_i.
- **Within** demeans within units $\rightarrow$ kills f_i, keeps f_t (next).
- **TWFE** does **both** — the destination of this lecture.

*Between removes f_t only in a **balanced** panel; "Within keeps f_t" means in demeaned form, $f_t - \bar{f}$.*

Section 2

Fixed effects

The within transformation

Subtract each unit's time-average from the model:

$$y_{it} - \bar{y}_i = \beta\left(x_{it} - \bar{x}_i\right) + \left(f_t - \bar{f}\right) + \left(u_{it} - \bar{u}_i\right).$$

f_i cancels — but f_t does not

f_i is constant over t, so demeaning subtracts it from itself ($f_i - f_i = 0$): the individual effect **disappears**, and $\hat{\beta}_{\text{within}}$ is consistent even if $\text{Cov}(x_{it}, f_i) \neq 0$. But the time effect leaves $f_t - \bar{f} \neq 0$ in the equation.

- OLS on the demeaned data **is** the **within / fixed-effects** estimator,

  $$\hat{\beta}_W = \frac{\sum_i \sum_t (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_i \sum_t (x_{it} - \bar{x}_i)^2},$$

  identified purely from **within-unit variation over time**.
- A common time shock can **still bias** it $\Rightarrow$ we add **time effects** (TWFE) next.

Within and between: one variance, split in two

The two estimators are not rivals — they read **two halves of the same variance**. For a balanced panel, the total variation in x decomposes **exactly**:

$$\underbrace{\sum_{it}(x_{it}-\bar{x})^2}_{\text{total}} = \underbrace{\sum_{it}(x_{it}-\bar{x}_i)^2}_{\text{within (over time)}} + \underbrace{T\sum_{i}(\bar{x}_i-\bar{x})^2}_{\text{between (across units)}} .$$

- **Within** uses only the first piece, **between** only the second — same data, two different questions.
- Pooled OLS uses **both**: it is the variance-weighted average

  $$\hat{\beta}_{\text{pooled}} = w\,\hat{\beta}_W + (1-w)\,\hat{\beta}_B, \qquad w = \frac{S_W}{S_W+S_B},$$

  with $S_W = \sum_{it}(x_{it}-\bar{x}_i)^2$ and $S_B = T\sum_i(\bar{x}_i-\bar{x})^2$ (the two pieces above — keep the T).
- **The price of within:** discarding the between piece ⇒ **larger standard errors**. Consistency is not free.

Balanced panel shown; unbalanced replaces T by T_i and weights accordingly.

Equivalence: least-squares dummy variables (LSDV)

Demeaning is identical to OLS with **one dummy per unit**: FE treats the unobserved f_i as **parameters to estimate** (one per unit), not as a random draw — each dummy's coefficient *is* f_i:

$$y_{it} = \beta x_{it} + f_i + u_{it}.$$

i	t	y_{it}	x_{it}	f_{USA}	f_{INDO}	f_{NIGER}
USA	2023	3.0	12	1	0	0
USA	2022	3.2	14	1	0	0
Indonesia	2023	1.0	22	0	1	0
Indonesia	2022	1.3	23	0	1	0
Niger	2022	0.1	28	0	0	1
Niger	2023	0.4	31	0	0	1

Same $\hat{\beta}$ as demeaning (Frisch–Waugh–Lovell). *Within* = looking **inside** each unit at its own evolution over time.

Frisch–Waugh–Lovell: within = partialling out f_i

Why are "demean" and "N dummies" the same answer? Collect the unit dummies in a matrix D and define the **residual-maker**

$$M_D = I - D(D'D)^{-1}D'.$$

- M_D is symmetric and idempotent[1] — a **projection**. Applied to any variable it **subtracts that variable's unit means**: M_D *is* the within transformation.
- **Frisch–Waugh–Lovell:** regressing y on (X, D) gives the **same** $\hat{\beta}$ as regressing the residualized $M_D y$ on $M_D X$:

$$\hat{\beta}_W = (X'M_D X)^{-1} X'M_D y.$$

- So we never invert the huge $N \times N$ dummy block — we demean and run OLS on k columns. This is exactly what `feols`/`reghdfe` do internally.

[1] Idempotent: applying it twice equals applying it once, $M_D M_D = M_D$.

What within can — and cannot — do

Strict exogeneity (the key assumption)

$\mathbb{E}[u_{it} \mid x_{i1}, \ldots, x_{iT}, f_i] = 0$: the shock is uncorrelated with *past, present and future* regressors. (Rules out feedback, e.g. a lagged y on the right-hand side.)

- ✓ Controls for **all** time-invariant confounders, observed or not — they sit in f_i.
- × **Cannot estimate** the effect of any **time-invariant** regressor: if $x_{it} = x_i$ then $x_{it} - \bar{x}_i = 0$ and it drops out (gender, distance, "was colonized"...).
- × Still **vulnerable to time-varying** confounders — those are *not* in f_i.

Section 3

Two-way fixed effects

Two-way fixed effects (TWFE)

The within estimator left $f_t - \bar{f}$ in the equation — a **common shock** each period (crises, world prices, technology) can still bias it. Add a **time effect** f_t beside the unit effect:

$$y_{it} = \alpha + \beta x_{it} + \underbrace{f_i}_{\text{unit}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

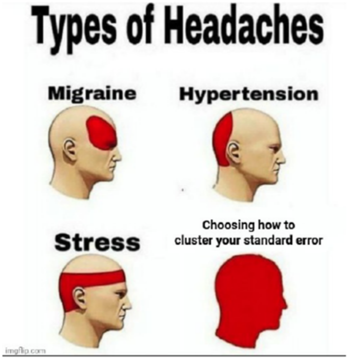
Estimate by **double-demeaning** ($\ddot{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$, same for y) — FWL with both dummy sets:

$$\hat{\beta}_{TWFE} = \frac{\sum_{it} \ddot{x}_{it}\, \ddot{y}_{it}}{\sum_{it} \ddot{x}_{it}^2}.$$

- Absorbs time-invariant **unit** heterogeneity **and** unit-invariant **time** shocks.
- Stricter control of confounders $\rightarrow$ the **workhorse** of applied micro and the basis of difference-in-differences.
- Identified only if $\ddot{x}_{it}$ still **varies** (not collinear with the time dummies) — else the denominator collapses.

Section 4

Clustered standard errors



Source: Khoa Vu

The variance of FE estimators: where inference enters

The first problem is settled: within and TWFE remove the effects from the error. The second remains: **what is Ω?** By FWL any FE estimator is OLS on M_D-residualized data, so we just **replace** X **by** $M_D X$ in the variance formula of the second-problem slide:

$$\mathrm{Var}(\hat{\beta}_{\mathrm{FE}}) = (X' M_D X)^{-1} \big(X' M_D \, \Omega \, M_D X\big) (X' M_D X)^{-1},$$

where M_D sweeps out the fixed effects — unit dummies for within, **both** sets for TWFE.

- If $\Omega = \sigma^2 I$ this collapses to the textbook $\sigma^2 (X' M_D X)^{-1}$ — what default software prints.
- But within a unit the shocks u_{it} are **serially correlated**, so Ω is **not** scalar and that textbook formula **understates** the true variance.
- The honest fix is to model Ω as block-diagonal by unit — **clustered standard errors**, the rest of this section.

Why i.i.d. standard errors fail

Default software sets $\Omega = \sigma^2 I$ — **all errors independent**, equal variance. The previous slide showed why that breaks; here is the intuition:

- A unit's shocks **persist** across its years, so its own observations are **not** independent — the off-diagonal of Ω is nonzero.
- *Analogy:* counting 200 students who copied off 3 as 200 independent answers — you *think* you have 200 observations; you have a handful.
- Consequence: SEs **too small** $\Rightarrow$ t-stats inflated $\Rightarrow$ you **over-reject** (the stars lie).

Clustering: count blocks, not observations

Allow **any** correlation *within* a cluster (a unit across its years) but independence *across* clusters $\Rightarrow \Omega$ is **block-diagonal**, and

$$X'\Omega X = \sum_{g=1}^{G} X_g' \, \Omega_g \, X_g.$$

- The **cluster-robust** estimator replaces each block by the residual outer product (Liang–Zeger); consistent as the **number of clusters** $G \to \infty$ — *not* as the periods per unit grow.
- So **effective information = number of clusters** G**, not** NT**.**
- **Moulton factor:** if the regressor varies at the cluster level, the ratio of the true to the naïve variance is $1 + (\bar{n} - 1)\rho$, where $\bar{n}$ is the **cluster size** (observations per cluster) and ρ the **within-cluster correlation** of the errors. With $\bar{n} = 50$, $\rho = 0.1$: the right SE is $\sqrt{6} \approx 2.5\times$ the naïve one.

Where — and how many — to cluster

- **Cluster at the level of treatment assignment** (or wherever residuals stay correlated). With repeated observations per unit, cluster **by the unit**, not unit–year — the errors are serially correlated over time (Bertrand et al. 2004).
- **Coarser is safer** (weakly larger SE) — but you need **enough clusters**.
- **Too few clusters** $\Rightarrow$ the formula breaks: a **wild-cluster bootstrap** (simulate the null instead of trusting the t-table) restores the test — see the *Going further* appendix.
- Default: with grouped/repeated data, **cluster by the group — and report** G**.**

Section 5

Application

A long-standing question

Does a hot climate lower economic growth?

> *There are countries where the excess heat … makes men lazy and discouraged.*
> *— Montesquieu (1748)*

The modern version: high temperatures reduce productivity and growth. How do we test it **credibly**? (Dell et al. 2012)

Cross-section / between: what is missing?

A cross-section (or between) regression of growth on temperature,

$$\text{Growth}_i = \gamma\,\text{Temp}_i + \varepsilon_i,$$

compares **hot vs. cold countries —** which differ in many *fixed* ways (f_i):

- hot, poor countries were often **colonized**, with lasting institutional damage;
- the industrial revolutions happened in **temperate** climates (accident + slow diffusion).

These are **time-invariant confounders** → the cross-section / between estimate is biased.

Within: temperature shocks inside each country

Add a country fixed effect and use **year-to-year temperature variation within each country**:

$$\text{Growth}_{it} = \beta\,\text{Temp}_{it} + f_i + u_{it}.$$

- f_i absorbs colonization, institutions, geography — *every* fixed country trait.
- Identification: does a country grow **less in its own hotter years**? (data: many countries, 1950–2003.)
- Still a threat: **time-varying** confounders correlated with temperature (El Niño cycles, adaptation) — a country fixed effect cannot remove those.

Results: no effect on average

	Annual growth rate
Temperature	-0.325
	(0.285)

Country fixed effects; SE clustered by country.

- Negative but **statistically insignificant**: temperature fluctuations do not move growth *on average across all countries*.
- But an average can hide **heterogeneity**…

Heterogeneity: poor vs. rich

- Poor countries are concentrated in **agriculture** (up to ~80% of activity), poorly insured.
- Temperature should bite **harder** there than in rich, service-based, air-conditioned economies.
- Test it with an **interaction**: let the temperature effect differ for poor countries.
- We also add **year effects** f_t — global shocks (recessions, oil prices) common to all countries would otherwise contaminate the estimate.

Interaction: the effect in poor countries

With both country and **year** fixed effects (f_i, f_t):

$$g_{it} = f_i + f_t + \beta\, T_{it} + \alpha\,(T_{it} \times POOR_i) + u_{it}.$$

The marginal effect of temperature **depends on** $POOR_i$ — so you **add** the two coefficients for poor countries:

$$\frac{\partial g_{it}}{\partial T_{it}} = \beta + \alpha\, POOR_i = \begin{cases} \beta & \text{rich } (POOR = 0) \\ \beta + \alpha & \text{poor } (POOR = 1). \end{cases}$$

	Annual growth rate
Temperature	0.261 (0.312)
Temperature × Poor	-1.655^{***} (0.485)

So the poor-country effect is $\beta + \alpha = 0.261 - 1.655 = -\mathbf{1.394}$ pp — vs ≈ 0 in rich countries.

Country & year FE; clustered by country. Testing $\beta + \alpha$ needs its ***own*** *SE (a linear-combination test), not the stars on α.*

Dell et al.: the clustering problem

The regression is a country-year panel. Its errors can be correlated in two directions.

- **Across years, within a country.** A growth shock persists, so the errors of one country in 1990 and 1991 share it. Clustering by country allows any correlation inside a country's block ($G = 125$).
- **Across countries, within a year.** A drought or an El Niño year hits neighbouring countries at once; the region$\times$year effect removes the common part, the rest is correlated inside the region-year cell. Clustering by region-year allows it.
- **Both at once: two-way clustering** (Cameron et al. 2011): country blocks plus region-year blocks, minus the overlap counted twice.

Dell et al. cluster by country in Table 2 (the estimates above) and two-way from Table 3 on: the practice of 2012. The next slide gives the current one.

Country-year panels today: two directions, two tools

Weather shocks are correlated over time inside a country and across space between countries. Each direction has its tool.

- **Serial correlation inside a country: cluster by country.** Dell et al. do this; 125 clusters is enough. It leaves the second direction untreated.
- **Spatial correlation between countries: Conley (1999) standard errors.** The correlation of shocks decays with distance, not at a region border. Each pair of observations enters Ω with a weight that declines with the distance between the units, up to a cutoff (1 000 km, say). A distance cutoff replaces the choice of a cluster. Standard in the climate–economy literature since Hsiang (2016).
- **Long panels: Driscoll–Kraay (1998) standard errors** allow any cross-sectional correlation but need a long T ($T \approx 50$ here). A cross-check of Conley.
- **Two-way clustering by country and region-year** has few groups in the second dimension; its variance can fail to be positive definite (MacKinnon et al. 2021). Prefer Conley when coordinates exist.
- **Whatever the choice, check the size.** Reassign the weather series across countries at random, re-estimate, count the rejections at 5 %: Dell et al. found 4 %. Randomisation inference remains the recommendation (Young 2019).

Section 6

Python

The toolkit and long format

Panel methods here use **`pyfixest`** — it *absorbs* fixed effects and clusters in one call (like Stata's `reghdfe` or R's `fixest`). Data must be in **long format**: one row per unit–period, with an `id` and a `time` column.

country	year	growth	temp
USA	2000	3.0	12
USA	2001	2.6	13
Niger	2000	0.5	28
Niger	2001	0.2	30

(Wide data — one column per year — must be **reshaped** to long first.)

Load the panel and check its structure

Four lines open the file and describe how many periods each unit is observed.

```
import pandas as pd, pyfixest as pf
df  = pd.read_csv("rica_panel.csv")   # 1
n_t = df.groupby("id").size()         # 2
df["id"].nunique()                    # 3
n_t.describe()                        # 4
```

- **1** — reads the file into a table `df` in long format: one row per farm and year.
- **2** — counts the rows of each farm, that is T_i. `groupby("id")` splits the table by farm, `.size()` counts the rows of each group; `n_t` has one value per farm.
- **3** — the number of distinct farms, N.
- **4** — the distribution of T_i: a minimum of 1 signals **singletons**, farms seen once, which carry no within variation and will be dropped by the fixed effects.

Within and between variation of a variable

Five lines split the variation of x into its within and between parts.

```
moy_i = df.groupby("id")["x"].transform("mean")     # 1
within = df["x"] - moy_i                            # 2
S_W = (within ** 2).sum()                           # 3
S_B = ((moy_i - df["x"].mean()) ** 2).sum()         # 4
w = S_W / (S_W + S_B)                               # 5
```

- **1** — the mean of x in each farm, written on **every row of that farm**; this is what `.transform("mean")` does, unlike `.mean()`, which would give one row per farm.
- **2** — each observation minus its farm's mean: the within deviation.
- **3** — the within sum of squares S_W.
- **4** — the between sum of squares S_B: each farm's mean minus the overall mean, squared and summed, so a farm counts T_i times.
- **5** — the within share w of the chapter. A small w means the estimator that uses within variation only works with a small part of the data.

The four estimators with one function

Seven lines estimate pooled, between, within and TWFE, with standard errors clustered by farm (line 1, stored once), and put them side by side.

```
cl = {"CRV1": "id"}                                          # 1
m_pool = pf.feols("y ~ x + C(region)", df, vcov=cl)          # 2
moy    = df.groupby("id")[["y", "x"]].mean()                 # 3
m_betw = pf.feols("y ~ x", moy, vcov="hetero")               # 4
m_with = pf.feols("y ~ x | id", df, vcov=cl)                 # 5
m_twfe = pf.feols("y ~ x | id + annee", df, vcov=cl)         # 6
pf.etable([m_pool, m_betw, m_with, m_twfe])                  # 7
```

- **2** — pooled OLS: all rows together, region dummies as controls.
- **3 and 4** — between: `moy` has one row per farm, the mean of `y` and `x`; the regression on these means uses cross-farm variation only.
- **5** — within: after `|`, `id` names the fixed effect to absorb, f_i.
- **6** — TWFE: `id + annee` absorbs f_i and f_t; singleton farms are dropped.
- **7** — the four results in one table, to watch $\hat{\beta}$ change as each effect enters.

Read the results and the fixed effects

Six lines pull out what the chapter asks for: coefficients, the year effects, residuals.

```
b, se = m_twfe.coef()["x"], m_twfe.se()["x"]      # 1
ci = m_twfe.confint().loc["x"]                    # 2
t  = (b - 0.136) / se                             # 3
fe = pd.Series(m_twfe.fixef()["C(annee)"])        # 4
fe.loc[2000] = 0.0                                # 5
u  = m_twfe.resid()                               # 6
```

- **1** and **2** — the coefficient, its standard error and its 95 % interval. In `pyfixest` these are **methods**, with parentheses, unlike `statsmodels` attributes.
- **3** — the t of $H_0 : \beta = 0{,}136$, a cost share; the summary tests zero only.
- **4** — the estimated year effects $\hat{f}_t$, a dictionary keyed by the fixed-effect name, turned into a series indexed by year.
- **5** — the reference year is absorbed and not returned; adding it as zero lets you read every $\hat{f}_t$ as a difference from 2000.
- **6** — the residuals, one per observation the estimation used, in the same order.

Four standard errors for one coefficient

Three lines re-estimate the same model under four assumptions on the error covariance.

```
for v in ["iid", "hetero", {"CRV1": "id"}, {"CRV1": "region"}]: # 1
    m = pf.feols("y ~ x | id + annee", df, vcov=v)               # 2
    print(m.se()["x"], m._G)                                      # 3
```

- **1** — the four choices of Ω from the appendix: classical, heteroskedastic, clustered by farm, clustered by region.
- **2** — `vcov` is the only thing that changes; $\hat{\beta}$ is identical each time.
- **3** — the standard error under each choice, and `_G`, the number of clusters the estimation actually used, after dropping singletons. The Moulton factor of the chapter is checked against the ratio of the third to the second.

Why cluster: the residuals of a farm resemble one another

Four lines measure the year-to-year correlation of a farm's residuals.

```
r = df_used[["id", "annee"]].copy()                  # 1
r["u"] = m_twfe.resid()                              # 2
r["u_lag"] = r.groupby("id")["u"].shift(1)           # 3
rho = r[["u", "u_lag"]].corr().iloc[0, 1]            # 4
```

- **1** — a small table with the farm and the year of every observation the estimation used; `df_used` is `df` without the singletons, so that it lines up with the residuals.
- **2** — attaches the residuals to it.
- **3** — the residual of the **previous row of the same farm**: `groupby("id")` keeps the shift inside each farm, `shift(1)` moves values down one row. Sort by farm and year first, and keep only consecutive years.
- **4** — the correlation between a residual and its lag, ρ. Positive means a farm's observations are not independent draws, which is what clustering accounts for.

References I

For a fuller textbook treatment of every method in this chapter, see Wooldridge (2025).

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *The Quarterly Journal of Economics* 119 (1): 249–75. https://doi.org/10.1162/003355304772839588.

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Section 7

Appendix

The covariance Ω: what we assume

Everything is the **sandwich** $\mathrm{Var}(\hat{\beta}) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$, with $\Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X]$ the $NT \times NT$ error covariance. Only the **meat** $X'\Omega X$ changes with our assumption:

Three choices for Ω — same $\hat{\beta}$, different SE

i.i.d.: $\Omega = \sigma^2 I$ (classical). **Heteroskedastic:** $\Omega = \mathrm{diag}(\sigma_i^2)$ (White/HC). **Clustered:** Ω block-diagonal (free within groups).

The point estimate never moves; the standard error is entirely a statement about Ω.

The cluster-robust estimator (Liang–Zeger, 1986)

With G clusters — errors free *within*, independent *across*:

$$X'\Omega X = \sum_g X_g'\Omega_g X_g \;\approx\; \sum_g X_g'\,\hat{\varepsilon}_g\hat{\varepsilon}_g'\,X_g,$$

$$\widehat{\mathrm{Var}}_{\mathrm{CR}}(\hat{\beta}) = \tfrac{G}{G-1}\,(X'X)^{-1}\Big(\textstyle\sum_g X_g'\hat{\varepsilon}_g\hat{\varepsilon}_g'X_g\Big)(X'X)^{-1}.$$

- Each cluster's residual outer product is a *poor* estimate of Ω_g — but the **sum over independent clusters** is consistent. $\frac{G}{G-1}$ is a small-sample fix.
- Consistent as $G \to \infty$ (**many clusters**), *not* as the cluster size grows.

Moulton (1990): the canonical lesson

A cluster-level regressor with equicorrelated errors (intraclass correlation ρ, cluster size $\bar{n}$) gives the **Moulton factor**:

$$\frac{\mathrm{Var}(\hat{\beta})_{\text{true}}}{\mathrm{Var}(\hat{\beta})_{\text{naive}}} = 1 + (\bar{n} - 1)\rho.$$

- $\bar{n} = 50, \rho = 0.1 \Rightarrow \sqrt{6} \approx 2.5\times; \bar{n} = 1000, \rho = 0.05 \Rightarrow \approx 7\times.$
- A country-level tax rate with 30{,}000 firms: information lives at the **country**, not the firm. You cannot escape clustering by disaggregating.

Bertrand–Duflo–Mullainathan (2004): the experiment

A famous stress-test of clustered inference. The question: do the **standard standard errors** used in policy evaluations actually have the right size?

- **Data.** The *Current Population Survey* (CPS) — the large monthly U.S. household survey — gives many years of individual wages across the 50 states.
- **Placebo law.** They invent a **fake** state policy: pick a random state and a random year, and pretend a law switched on there. It changes nothing, so the true effect is **zero by construction** ($\beta = 0$).
- **The test.** Estimate the "effect" of this fake law on wages and ask how often the t-test rejects $\beta = 0$ at the 5% level. With honest SEs it should reject $\approx 5\%$ of the time. Repeat over hundreds of random placebos.

Bertrand–Duflo–Mullainathan (2004): the trap, and the fix

The result

Classical (i.i.d.) SEs rejected the true null in $\approx 45\%$ of placebos — nine times the nominal 5%. Clustering by **state** brought it back to 6–8%.

- **Why so bad?** Wages are **serially correlated** within a state (this year looks like last year), *and* the fake law, once on, **stays on** for years (treatment is persistent). The many state–year rows are far from independent — effectively there is only **one experiment per state**, not one per state–year.
- **The fix.** Cluster at the **state** level (the unit of treatment): the SEs then count ≈ 50 blocks, not thousands of rows, and the test size is restored.
- **Lesson.** Cluster at the level of treatment assignment — not the finer unit–year.

Abadie–Athey–Imbens–Wooldridge (2023): when to cluster, really

Reframes clustering as a question about **design**, not residuals. Cluster if **either**:

1. **Sampling design —** you sampled clusters, then units within them (schools, then pupils).
2. **Assignment design —** treatment is assigned at the cluster level.

- If **neither** holds (a census; unit-specific treatment), clustering may be unnecessary and can even *cost* precision.
- “Are errors correlated?” is the wrong question; “did the data or the treatment arrive in clusters?” is the right one.

Where to cluster: the practical rule

Rule

Cluster at the **coarsest** level at which (i) treatment is assigned / sampling was done, or (ii) residuals stay correlated.

- Nested clusterings $C_1 \subset C_2$: $\widehat{\mathrm{Var}}_{C_2} \geq \widehat{\mathrm{Var}}_{C_1}$ — **coarser is (weakly) larger**, hence safer for test size.
- But coarser shrinks G; below $\sim$ 15–20 clusters the formula degrades.
- **Never cluster finer than the level of treatment assignment.**

Multiway clustering (Cameron–Gelbach–Miller, 2011)

Correlation along **two** non-nested dimensions (e.g. country *and* year):

$$\widehat{\text{Var}}_{\text{2-way}} = \widehat{\text{Var}}_{\text{country}} + \widehat{\text{Var}}_{\text{year}} - \widehat{\text{Var}}_{\text{country} \cap \text{year}}.$$

- Each piece is a valid one-way CRV; subtracting the intersection avoids double-counting.
- Use only with a clear reason for two dimensions; **unstable** when one dimension has few clusters (e.g. few years).

Few clusters: two distinct problems

CRV1 is consistent only as $G \to \infty$. When G is small it misbehaves in **two** ways:

1. **Reference distribution.** Compare t to t_{G-1}, not the normal (e.g. $t_{21,\,0.975} = 2.08$ vs 1.96) — software does this automatically.
2. **Noisy variance.** $\sum_g X_g' \hat{\varepsilon}_g \hat{\varepsilon}_g' X_g$ is itself a noisy estimate and **under-covers**, worse with unbalanced clusters.

Each needs a *different* remedy.

Few clusters: the remedies

Rather than trust the t-table when G is small, **simulate** the test's own distribution and read the p-value off it.

- **Wild-cluster bootstrap** (Cameron–Gelbach–Miller 2008): re-estimate the model thousands of times, each time flipping the sign of every cluster's residuals at random (± 1). This builds the distribution of the t-statistic *under the null*, using only G blocks. The historic default — but it **fails with few *treated* clusters** (can trap $p \approx 1$). Use with care.
- **CR2 + Satterthwaite df** (Imbens–Kolesár 2016; Pustejovsky–Tipton 2018): bias-corrected sandwich + effective df. Good default to $G \approx 10$ when leverage is balanced.
- **CR3 cluster jackknife** (MacKinnon–Nielsen–Webb 2023): robust when a few clusters carry most of the leverage.
- **Randomization inference** (Young 2019): permute treatment across clusters; distribution-free.

Summary, and further reading

Decision tree

Assigned/sampled in clusters, or residuals correlated? $\rightarrow$ cluster there. Coarser $=$ safer, but keep enough clusters. Few clusters $\rightarrow$ CR2 / CR3 or randomization inference, **not** a naive wild bootstrap when treatment is concentrated.

Moulton (1990); Liang–Zeger (1986); Bertrand–Duflo–Mullainathan (2004); Cameron–Gelbach–Miller (2008, 2011); Cameron–Miller (2015); Imbens–Kolesár (2016); Pustejovsky–Tipton (2018); Abadie–Athey–Imbens–Wooldridge (2023); MacKinnon–Nielsen–Webb (2023); Young (2019).